Do Now · retrieval

Five process scheduling algorithms

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a Ready process?

Q2

What is a scheduling quantum?

Q3

What is a context switch?

Q4

Sorted [1,3,5,7,9,11,13,15,17], target 15. Trace midpoint indexes.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — FCFS serves arrival order; SJF selects the shortest available complete job.

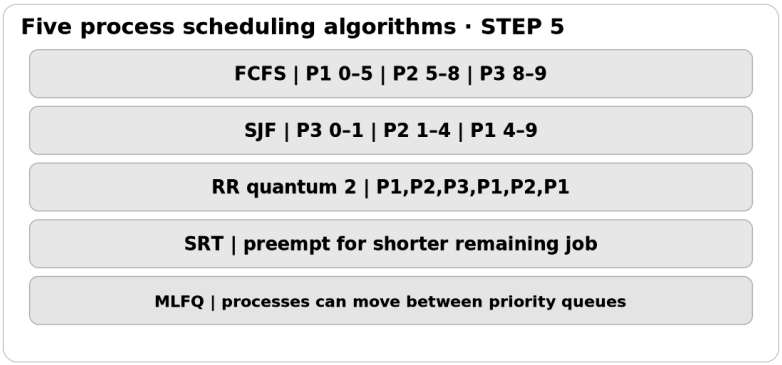
Core idea 2

Trace — SRT can preempt when a newly ready job has shorter remaining time; RR cycles by quantum.

Core idea 3

Evaluate — Multilevel feedback queues adapt priority by task behaviour. Compare responsiveness, waits, starvation and context-switch overhead.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

FCFS — First Come First Served.

SJF — Shortest Job First.

SRT — Shortest Remaining Time.

RR — Round Robin.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A=4,B=2 arrive together: trace non-preemptive SJF.

Follow the worked method

1. Predict first — Shorter job B runs first

2. Model and check — A waits two time units.

Source model · final illustrated step

IDO · STEP 2

Shorter job B runs first

A waits two time units.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
Arrival order A,B,C gives mean waiting (0+3+5)/3=8/3.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Name FCFS,SJF,SRT,RR and multilevel feedback by defining feature. [5]

Question 2

A=5,B=3,C=1 all arrive at t=0. For FCFS, break ties in A, B, C order; for non-preemptive SJF, select the shortest available job. Compare the two orders and mean waiting times. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Using Round Robin with time quantum 2, processes A=4, B=3 and C=1 all arrive at t=0; take the initial queue order as A, B, C. Give all execution slices and one drawback of a short quantum. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Round Robin always lets one process finish before any other runs.

Use key vocabulary from memory

1. Explain FCFS in your own words.

2. Explain SJF in your own words.

3. Explain SRT in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.